CS BS

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[23CSBSMA21]

# SECOND SEMESTER B.Tech EXAMINATION, JUNE - 2024

## STATISTICAL METHODS AND MODELLING

**Time: 3 Hours** **Maximum Marks: 100**

**Instructions:** i. Missing data may be suitably assumed.

**ANSWER ALL QUESTIONS**

| | | | | | |
|---|---|---|---|---|---|
| | | **PART - A** | | **05x02=10** | |
| **1.** | **a.** | Illustrate non-probabilistic sampling with an example. | L2 | **CO1** | **02** |
| | **b.** | If $r_{12} = 0.77, r_{13} = 0.72, r_{23} = 0.52$, then find $r_{12.3}$. | L3 | **CO2** | **02** |
| | **c.** | Discuss the two tailed test. | L2 | **CO3** | **02** |
| | **d.** | Given<br>X: 60 45 23 32<br>Y: 10 25 20 54 32 65 8.<br>Apply Mann Whitney test and find the sum of ranks of Y values. | L2 | **CO4** | **02** |
| | **e.** | List the components of Time Series Analysis. | L1 | **CO5** | **02** |
| | | **PART - B** | | **06x05=30** | |
| **2.** | **a.** | The weight of 1500 ball bearings is normally distributed with mean of 635 gm and standard deviation of 1.36 gm. If 300 sample of size 36 are drawn from population, calculate the mean and standard deviation of sampling distribution of means if sampling is done<br>**i.** with replacement<br>**ii.** without replacement. | L2 | **CO1** | **05** |
| | | **OR** | | | |
| | **b.** | Describe the following.<br>**i.** Estimate<br>**ii.** Estimator<br>**iii.** Sample mean<br>**iv.** Point estimation<br>**v.** Population mean. | L2 | **CO1** | **05** |
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| **3.** | **a.** | Fit a straight line for the following data<br>X: 5, 10, 15, 20, 25<br>y: 16, 19, 23, 26, 30<br>Hence estimate the value of $y$ corresponding to $x = 30$ . | L2 | **CO2** | **05** |
| | | **OR** | | | |
| | **b.** | Show that if $\theta$ is the angle between the lines of regression, then $tan\,\theta = \frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2}\left(\frac{1-r^2}{r}\right)$. | L2 | **CO2** | **05** |
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| **4.** | **a.** | A random sample of size $n_1 = 25$ taken from a normal population with standard deviation $\sigma_1 = 5.2$ has a mean $\bar{x}_1 = 81$. A second random sample size of $n_2 = 36$ taken from different normal population with standard deviation $\sigma_2 = 3.4$ has a mean $\bar{x}_2 = 76$. Discuss the hypothesis that $\mu_1 = \mu_2$ against $\mu_1 > \mu_2$ at 5% level of significance.(given $Z_\alpha = 1.645$) | L3 | **CO3** | **05** |
| | | **OR** | | | |

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|---|---|---|---|---|---|
| | b. | The mean mathematics score for the population of first-year students at University is 520. The standard deviation of scores in this population is 95. An investigator believes that the mean mathematics score of first-year B.Tech course is significantly different from the mean score of the population. The mean of a sample of 36 first year B.Tech students is 548. Test the investigator's prediction using an alpha level of 5%.(Z-value:1.96) | L3 | **CO3** | **05** |
| | | | | | |
| **5.** | **a.** | Implement Sign test for testing of hypothesis at 5% level of significance. Following are the responses to the question "How many hours do you study before a major statistics test?"<br>6 5 1 2 2 5 7 5 3 7 4 7<br>Given that the median number of hours a student studies before a test is 3. Critical value for sign test for n=11 is 1 for two tailed test. | L3 | **CO4** | **05** |
| | | **OR** | | | |
| | **b.** | Test the hypothesis that the median HDL cholesterol levels in adult population city A and city B are same. Using the following observation and the Mann-Whitney test at 5% level of significance by<br>City A 42 20 51 39 57 60 23<br>City B 30 42 25 29 35<br>Given that, the critical value of U for $n_1 = 7$, $n_2 = 5$ at 5% level of significance for two tailed is 5. | L3 | **CO4** | **05** |
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| **6.** | **a.** | The net profits of a business have been given below, examine net profit for 2012 by least square trend method. | L3 | **CO5** | **05** |

| Years | 2006 | 2007 | 2008 | 2009 | 2010 | 2011 |
|---|---|---|---|---|---|---|
| $ | 35 | 40 | 38 | 45 | 53 | 60 |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | **OR** | | | | | | | |
| | **b.** | Compute 5 year moving averages centred for this time series data | L3 | **CO5** | **05** | | | | |

| Year | 2001 | 2002 | 2003 | 2004 | 2005 | 2006 | 2007 | 2008 | 2009 |
|---|---|---|---|---|---|---|---|---|---|
| Sales | 13 | 15 | 17 | 18 | 19 | 20 | 20 | 21 | 22 |

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| **7.** | **a.** | Write R-program to create a table for the following data frames. | L3 | **CO3** | **05** |

| Names | Maths | Physics | Chemistry |
|---|---|---|---|
| Kavya | 90 | 70 | 95 |
| Chandan | 80 | 80 | 65 |
| Ramya | 40 | 60 | 86 |
| Sahana | 30 | 90 | 75 |

| | | | | | |
|---|---|---|---|---|---|
| | | **OR** | | | |
| | **b.** | Following are the weights of four bags of rice(in kgs) chosen at random from a lot of 100 bags: | L3 | **CO1** | **05** |

| | | | |
|---|---|---|---|
| 101kg | 99kg | 98kg | 102kg |

Find the best estimator of true mean weight, variance and standard deviation of all the bags.

| | | PART - C | | | 06x10=60 |
|---|---|---|---|---|---|
| **8.** | **a.** | A population consists of the four numbers 3, 7, 11, 15. Consider all possible samples of size 2 which can be drawn with replacement. Estimate<br>**i.** Population Mean and population Standard Deviation.<br>**ii.** Mean and Standard Deviation of the sampling distribution of means. | L3 | **CO1** | **10** |
| | | **OR** | | | |
| | **b.** | For a random sample $x_1, x_2, \ldots, x_n$ assume that $x_i$'s are independent Bernoulli random variable of the students selecting a course of statistics with unknown parameters P. Calculate the Maximum likelihood estimator of P. | L3 | **CO1** | **10** |
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| **9.** | **a.** | Fit a second degree parabola $y = ax^2 + bx + c$ in the least square sense for the following data and hence estimate $y$ at $x = 6$.<br>x: 1, 2, 3, 4, 5<br>y: 10, 12, 13, 16, 19 | L3 | **CO2** | **10** |
| | | **OR** | | | |
| | **b.** | Obtain the lines of regression for the data. Hence find the coefficient of correlation for the data.<br>$x$: 1, 3, 4, 2, 5, 8, 9, 10, 13, 15<br>$y$: 8, 6, 10, 8, 12, 16, 16, 10, 32, 32 | L3 | **CO2** | **10** |
| | | | | | |
| **10.** | **a.** | Can the following two samples can be regarded as coming from same normal population?<br>Sample 1: Size 10, Sample mean 12, sum of squared deviation from mean 120<br>Sample 2: Size 12, Sample mean 15, sum of squared deviation from mean 314<br>{given $((F_{0.05}(11,9) = 3.10)$} | L3 | **CO3** | **10** |
| | | **OR** | | | |
| | **b.** | A sample of 100 bulbs produced by a company A showed a mean life of 1190 hours and a Standard Deviation of 90 hours. Also a sample of 75 bulbs produced by a company B showed a mean life of 1230 hours and a Standard Deviation of 120 hours. Is there a difference between the mean life time of the bulbs produced by the two companies at 5% and 1% level of significance (LOS). {given (for 5% LOS $, Z_{\alpha/2} = \mp 1.96$). (For 1% LOS $Z_{\alpha/2} = \mp 2.58$).} | L3 | **CO3** | **10** |
| | | | | | |
| **11.** | **a.** | Following table shows the frequency distribution of college student according to their pocket money (daily). Write R-programing commands to construct histogram and frequency polygon on the same graph paper.<br>Pocket Money (In Rs): 20-29, 30-39, 40- 49, 50-59, 60-69, 70-79, 80-89<br>No. of Students: 10, 24, 18, 12, 8, 5, 3 | L3 | **CO4** | **10** |
| | | **OR** | | | |

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|---|---|---|---|---|---|
| | **b.** | The following data gives the HDL and LDL cholesterol levels of 7 adults in a locality. | L3 | **CO4** | **10** |

| HDL(x) | 36 | 39 | 23 | 31 | 33 | 51 | 45 |
|---|---|---|---|---|---|---|---|
| HDL(y) | 80 | 72 | 101 | 90 | 98 | 70 | 50 |

Obtain the rank correlation coefficient between the HDL and LDL cholesterol level.

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| **12.** | **a.** | Compute 5-year, 7-year and 9-year moving averages for the following data: | L3 | **CO5** | **10** |

| year | 1990 | 1991 | 1992 | 1993 | 1994 | 1995 | 1996 | 1997 | 1998 | 1999 |
|---|---|---|---|---|---|---|---|---|---|---|
| values | 2 | 4 | 6 | 8 | 10 | 12 | 14 | 16 | 18 | 20 |

**OR**

| | | | | | |
|---|---|---|---|---|---|
| | **b.** | Discuss Time Series its application in forecasting by taking any real time application. | L3 | **CO5** | **10** |
| **13.** | **a.** | A test was given to five students taken at random from the fifth class of three school of a town. The individual scores are. | L3 | **CO6** | **10** |

| School-I | 9 | 7 | 6 | 5 | 8 |
|---|---|---|---|---|---|
| School-II | 7 | 4 | 5 | 4 | 5 |
| School-III | 6 | 5 | 6 | 7 | 6 |

Carry out the analysis of variance and write your analytical inference on the result.

**OR**

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|---|---|---|---|---|---|
| | **b.** | Fit a curve of the form $y = ae^{bx}$ for the data using least square method. | L3 | **CO2** | **10** |

| $x$ | 0 | 2 | 4 |
|---|---|---|---|
| $y$ | 8.12 | 10 | 31.82 |